

## Quadratic Transformations &amp; Characteristics

Date \_\_\_\_\_ Period \_\_\_\_\_

**Identify the vertex and axis of symmetry of each. Describe the transformation from the parent function.**

1)  $f(x) = (x - 1)^2 + 9$

2)  $f(x) = -4(x + 10)^2 + 10$

3)  $f(x) = 2(x + 9)^2$

4)  $f(x) = -9(x - 2)^2 - 6$

5)  $f(x) = -\frac{1}{2}(x - 2)^2 - 2$

6)  $f(x) = (x + 2)^2 - 6$

**Identify the vertex, axis of symmetry, and y-intercept of each. Then sketch the graph. Determine the domain, range, and interval of increase and decrease.**

7)  $y = x^2 - 8x + 10$

8)  $y = x^2 - 2$

9)  $y = -(x - 2)^2$

10)  $y = -\frac{1}{4}(x - 3)^2 + 2$

# Answers to Quadratic Transformations & Characteristics

1) Vertex:  $(1, 9)$   
Axis of Sym.:  $x = 1$

2) Vertex:  $(-10, 10)$   
Axis of Sym.:  $x = -10$

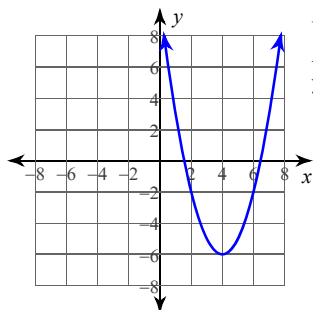
3) Vertex:  $(-9, 0)$   
Axis of Sym.:  $x = -9$

4) Vertex:  $(2, -6)$   
Axis of Sym.:  $x = 2$

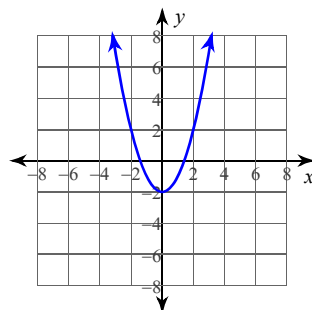
5) Vertex:  $(2, -2)$   
Axis of Sym.:  $x = 2$

6) Vertex:  $(-2, -6)$   
Axis of Sym.:  $x = -2$

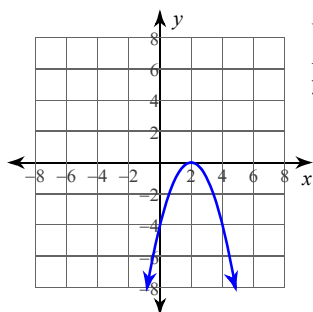
7) Vertex:  $(4, -6)$   
Axis of Sym.:  $x = 4$   
y-int: 10



8) Vertex:  $(0, -2)$   
Axis of Sym.:  $x = 0$   
y-int: -2



9) Vertex:  $(2, 0)$   
Axis of Sym.:  $x = 2$   
y-int: -4



10) Vertex:  $(3, 2)$   
Axis of Sym.:  $x = 3$   
y-int:  $-\frac{1}{4}$

